

Suppression of Magnetic Noise and Field in Cubic Low-Noise Ferrite Magnetic Shields

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Abstract—Most magnetic shielding rooms (MSRs) are made of highly permeable permalloy. The limitation of such MSR is its intrinsic magnetic noise, which exceeds the sensitivity of magnetic sensors, impeding the further applications of ultra-sensitive magnetic sensors, such as the magnetometers operated in the spin-exchange relaxation free (SERF) regime. In this article, a cubic ferrite magnetic shield with a low magnetic noise and magnetic field is proposed. The demagnetization model and magnetic noise calculation models are given. After optimizing several different coil structures and degaussing parameters, the magnetic field and the magnetic noises inside the cubic ferrite shield were measured. Within a distance of ± 10 mm near the center, the residual magnetic field is below 0.5, 1.72, and 1.57 nT at the x -, y - and z -directions. The magnetic noise measurements at the center of the cubic ferrite magnetic shield reveal an averaged magnetic noise of $2.62 \text{ fT/Hz}^{-1/2}$ from 20 to 100 Hz. Compared with the noise spectrum of permalloy shield, the noise level is reduced more than 133%. The results of this scaled-down cubic ferrite magnetic shield can be applied in larger sized shielding rooms. By assembling multiple ferrite panels together, it is possible to create a larger shielding space with a low magnetic field and low magnetic noise.

Index Terms—Demagnetization, hysteresis loss, magnetic noise, magnetic shields, spin-exchange relaxation free (SERF) magnetometer.

I. INTRODUCTION

MAGNETIC shielding rooms (MSRs) can provide a very low magnetic field environment and have been widely

used in the domains of electrical engineering, such as wireless charging system [1], power transfer [2], magnetoencephalography [3], [4], fundamental physics research [5], and industrial applications [6].

In most MSRs, the magnetic flux shunt effects of highly permeable materials, such as permalloy and ferrites, can reduce the external static magnetic interferences and create a low field environment for the measuring cite [7], [8], [9]. For example, the most promising MSR made of permalloy has achieved a magnetic field less than 0.8 nT in 1-m^3 area [10], [11], [12].

Although the external electromagnetic interferences can be reduced by using the magnetic shield, the dissipative material made the magnetic shield produce the hysteresis magnetic noise and eddy current magnetic noise, which both limit the performance of many industrial electronics, especially of the magnetometers operated in the spin-exchange relaxation free (SERF) regime [13]. Previous studies show that the direct magnetometers' measurement inside the permalloy magnetic shield reveals a sensitivity of $10 \text{ fT/Hz}^{-1/2}$, mainly limited by the eddy current magnetic noise generated by the permalloy [14]. To date, permalloy [15], thin aluminum/steel sheets [16], coppers [17], and other novel metallic materials [18] are widely used to make MSRs. The drawback of these materials is that their high electrical conductivities inevitably generate a large eddy current magnetic noise, impeding the further applications of ultrasensitive magnetic sensors.

As shown in Table I, the highly permeable ferrites are promising for magnetic noise reduction due to their much lower electrical conductivity than that of the permalloy [19]. Kornack et al. [20] derived the magnetic noise of a cylindrical ferrite magnetic shield based on the generalized Nyquist relation and realized a magnetic noise level of $0.75 \text{ fT/Hz}^{-1/2}$, a level 25 times lower than that of a permalloy magnetic shield with a similar size. This noise level can be further reduced by using a gradiometer arrangement inside a cylindrical ferrite magnetic shield. Dang et al. [21] demonstrated an SERF magnetometer with a gradient sensitivity of $0.16 \text{ fT/Hz}^{-1/2}$, which is the highest sensitivity in record. To further improve the sensitivity, the magnetic noises of ferrites were investigated. Studies have correlated the magnetic noise with the microstructures of ferrites and found that a larger grain size is favorable for reducing the hysteresis magnetic noise [22]. Ferrite magnetic shields are effective in reducing the magnetic noise [20], [23]. However, current processing technologies

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TABLE I
PROPERTIES OF MATERIALS EMPLOYED IN MAGNETIC SHIELDING

Materials	μ_m	Densit y (g/cm ³)	B_s	σ (mS/m)	H_c (A/m)
Mn-Zn ferrites [24]	2200	4.9	530 mT	250	13
Permalloy [25]	75000	8.74	750 mT	1.7×10^9	1.2
Amorphous alloys [26]	146000	-	500 mT	2.2×10^8	1.6
Soft magnetic composites [27]	38	5.95	169 emu/g	7.5×10^4	780

of ferrites cannot create the enclosure, whose geometry and volume are equivalent to that of MSRs.

This article demonstrates a low-noise cubic ferrite magnetic shield assembled by several Mn-Zn ferrite panels. The biggest advantage of this geometry is that it can be employed in the MSRs, serving as the innermost layers to reduce the large eddy current magnetic noise generated by the permalloys. This article is organized as follows. First, the theoretical models of magnetic noise calculation and demagnetization are described (Section II). Then, the magnetic field inside the ferrite magnetic shield is simulated, and demagnetization is optimized by considering different coil structures (Section III). After the demagnetization, magnetic field and magnetic noises inside the cubic ferrite shield were measured (Section IV). The results of this scaled-down cubic ferrite magnetic shield can be applied in the larger sized shielding rooms. By assembling multiple ferrite panels together, it is possible to create a larger shielding space with a low magnetic field and low magnetic noise.

II. MODELING AND ANALYSIS

A. Magnetic Noise of Cubic Ferrite Magnetic Shielding

The magnetic noise can be calculated from the power loss in the source materials [28]. In the magnetic shield, the magnetic noise at any point is due to the dissipation power of materials. According to Faraday's law, a pickup coil will develop a fluctuating voltage when there was a magnetic field fluctuation with a power spectral density $S_B(f)$

$$S_V(f) = A^2 N^2 \omega^2 S_B(f) \quad (1)$$

where A is the area of coil, N is the number of turns, and ω is the angular frequency. Meanwhile, the fluctuation dissipation theorem states that the fluctuating voltage of a pickup coil has the following relation:

$$S_V(f) = 4kTR_{\text{eff}}(f) \quad (2)$$

where T is the temperature in Kelvin, k is Boltzmann's constant, and R_{eff} is the effective resistance. Considering the time averaged power dissipation $P(f)$ incurred by an oscillating current $I(t) = I \sin \omega t$

$$P(f) = \frac{1}{2} I^2 R_{\text{eff}}(f) \quad (3)$$

the magnetic field noise equation can be derived from (1)–(3)

$$\delta B(f) \equiv \sqrt{S_B(f)} = \frac{\sqrt{4kT} \sqrt{2P(f)}}{ANI\omega}. \quad (4)$$

In a linear response regime, δB is independent of A and I because the power dissipation $P(f)$ scales with the driving dipole $p \equiv ANI$ [28]. The usefulness of (4) is it can characterize the magnetic field noise due to different sources. For highly permeable ferrites used in this study, the primary sources of power dissipation are eddy current power loss and hysteresis loss, whose expressions are shown as follows:

$$P_{\text{eddy}} = \int_V \frac{1}{2} \sigma E^2 dV \quad (5)$$

where σ is the electrical conductivity

$$P_{\text{hyst}} = \int_V \frac{1}{2} \omega \mu'' H^2 dV \quad (6)$$

where μ'' is the imaginary part of complex permeability, σ is the electrical conductivity, E is the electric field intensity, and H is the magnetic field strength. According to (5), because the electrical conductivity of ferrites is 10^6 times smaller than that of permalloy [13], the magnetic noise due to eddy current power loss can be effectively suppressed by using the ferrite magnetic shield. Because the ferrite magnetic shield has a complex geometry, finite element method is used to analyze the magnetic field strength H [29]. According to previous studies, at the low frequencies (<100 Hz), the primary magnetic noise of ferrite magnetic shield is the hysteresis loss [13]. Therefore, the magnetic noise due to the hysteresis power loss is mainly analyzed in this article. By putting (6) into (4), the magnetic noises can be expressed as

$$\delta B \approx \delta B_{\text{hyst}} = \frac{\sqrt{2kT\mu''}}{AI\sqrt{\pi f}} \left(\int_V H^2 dV \right)^{1/2}. \quad (7)$$

Because the cubic magnetic shield is centrally symmetric, the longitudinal noise is equivalent to transverse ones at the center.

Using the calculation model above, the magnetic noises of ferrite magnetic shield were analyzed using the ANSYS Maxwell software. In finite element analysis, the cubic ferrite magnetic shield is placed inside six-layered permalloys. Such magnetic shield combined the innermost ferrite and outer permalloy has been proved to suppress the magnetic noise to sub fT/Hz^{-1/2} [29], [30], [31]. To analyze the magnetic field strength H over the volume, a toroidal coil with an outer diameter of 5 mm and an inner diameter of 0.5 mm is placed at the center of the cubic ferrite magnetic shield. The coil carrying an excitation current of 1 A introduced a power loss to the ferrites. Because the ferrite material has a high resistivity ($\sim 1 \Omega \cdot \text{m}$), the eddy current effect is ignored when the excitation current is small and the frequencies are below 100 Hz [13]. The power loss of cubic ferrite magnetic shield is mainly due to the magnetic hysteresis loss. The magnetic noise can be calculated when the integral of magnetic field strength H^2 over the volume is determined, as written in (7).

According to (7), the magnitude of the magnetic noise depends on the integral of magnetic field strength H^2 over the volume. Since the ferrite magnetic shield has holes on it, $\int H^2 dV$ needs to be analyzed using the ANSYS Maxwell software. In this work, the maximum side length of ferrite

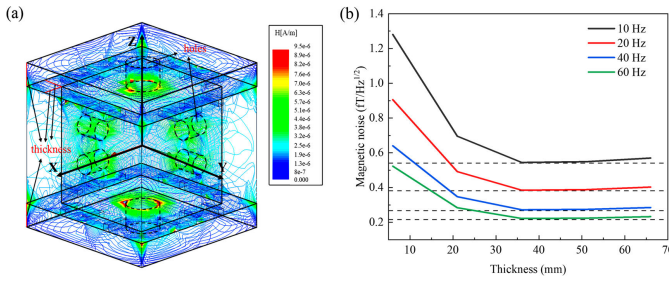


Fig. 1. Optimization of the wall thickness. (a) Simulation model. (b) Calculated magnetic noise with different wall thicknesses at 25 °C. Dashed lines represent the lowest noise level at each frequency.

shield is 240 mm, limited by the load of dry pressing in the ferrite processing. With a max side length of 240 mm, the optimized wall thickness was obtained when it yields the smallest value of $\int H^2 dV$, as shown in Fig. 1(a). The wall thickness is dependent on the side length, and the finite element analysis is required to obtain the best thickness to side length ratio for different lengths. In this work, the magnetic noises of ferrite magnetic shields with different wall thicknesses were analyzed, as shown in Fig. 1(b). When the frequency is the same, the magnetic noise decreases with the wall thickness and all reaches the lowest level at 36 mm. Further increase in the wall thickness will increase the magnetic noise. Based on this result, when the side length of the cubic ferrite magnetic shield is constant (240 mm), a wall thickness of 36 mm yields the lowest noise level. Therefore, actual ferrite magnetic shield with a wall thickness of 36 mm is fabricated, as shown in Fig. 2(a) and (b), and its performance was further investigated in the later experiment.

B. Demagnetization Model of High Permeability Ferrites

Due to the hysteresis characteristics of ferrite materials, the residual magnetization presents in the magnetic shield after demagnetization. The residual magnetization can be added to the target signals and deteriorates magnetometers' measuring accuracy. Thus, a high-performance magnetic shield requires a minimum residual magnetization. In this study, dynamic Jiles–Atherton (JA) model is integrated with finite element method to calculate theoretical residual magnetization in the cubic ferrite magnetic shield [6], [31], [32]. The JA model is expressed as follows:

$$\frac{\partial \vec{M}}{\partial t} = (1 - c) \frac{\partial \vec{M}_{\text{irr}}}{\partial t} + c \frac{\partial \vec{M}_{\text{an}}}{\partial t} \quad (8)$$

$$\vec{H}_e = \vec{H} + \alpha \vec{M} \quad (9)$$

$$\vec{M}_{\text{an}} = M_s \left(\coth \frac{|\vec{H}_e|}{a} - \frac{a}{|\vec{H}_e|} \right) \frac{\vec{H}_e}{|\vec{H}_e|} \quad (10)$$

$$\frac{\partial \vec{M}_{\text{irr}}}{\partial t} = g \left(\frac{\vec{M}_{\text{rev}}}{ck} \frac{\partial \vec{H}_e}{\partial t} \right) \frac{\vec{M}_{\text{rev}}}{|\vec{M}_{\text{rev}}|} \quad (11)$$

$$\vec{M}_{\text{rev}} = c (\vec{M}_{\text{an}} - \vec{M}_{\text{irr}}) \quad (12)$$

where magnetization M contains an irreversible magnetization M_{irr} and a reversible magnetization M_{rev} , M_{an} is the

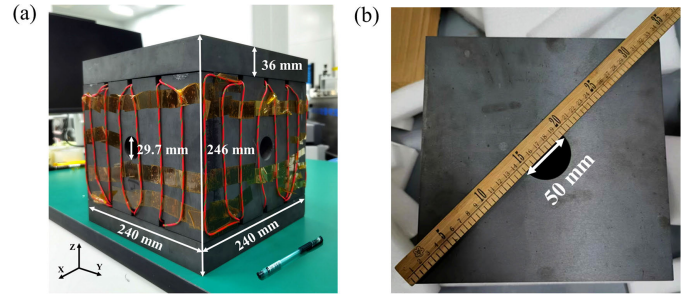


Fig. 2. Photograph of cubic ferrite magnetic shield. (a) Dimetric view. (b) Top view.

anhysteretic magnetization, and M_s is the saturation magnetization. α is the mean domain coupling parameter, a is the density of domain wall, c is the reversibility for magnetization, g is a function depends on the magnetization pathway x , $g(x) = 1$ for $x \geq 0$ and otherwise $g(x) = -1$, and k is the required energy to break the pinning site. A promising demagnetization requires a magnetic field H larger than the saturated magnetic field to be applied to the cubic ferrite magnetic shield, and the applied magnetic field is expressed as a polynomial envelope function [10]

$$H(t) = H_o(1 - t/T)^n \sin(2\pi ft) \quad (13)$$

where H_o is the initial magnetic field, t is the time, T is the duration for sine function approaching to zero ($t \leq T$), and n is the order to polynomial ($n = 1$ or $n = 2$).

JA model uses the parameters, such as M_s , α , a , c , g , and k , to calculate the magnetic flux density at different magnetic fields H . There are six unknown parameters in five equations, so the analytical solutions for JA model do not exist. MATLAB software is used to calculate the numerical solutions of JA models. First, the initial values for M_{rev} and M_{irr} are set to 0. For a small change in the magnetic field dH , dM_{irr} is determined using (11), and new irreversible magnetization M_{irr} will be the sum of initial M_{irr} and dM_{irr} . By putting M_{irr} into (12), M_{rev} can be determined. Finally, the anhysteretic magnetization M_{an} and effective magnetic field H_e can be calculated from M_{irr} and M_{rev} . To determine the initial value of M_s , the static hysteresis loop of ferrite is measured using MATS-2010SD analyzer (Hunan Linkjoin Technology). In Fig. 3, the result of the hysteresis loop indicates that the saturation magnetization M_s of ferrite materials is 2.96×10^5 A/m, which is approximately 0.37 T according to $B = \mu_0(H + M)$, and its coercivity is 1.4 A/m. This measured saturation magnetization is used as the initial approximation for M_s in the JA model.

During the demagnetization, a magnetic field is applied to the ferrite materials and the hysteresis loop $B(H)$ will circle down to $B = 0$. The remanence at the endpoint is the residual magnetization. Previous studies have shown that a lower residual magnetization could be obtained by applying a magnetic field with a higher order envelope function [31]. Here, a magnetic field with a second-order envelope function is applied to the ferromagnetic materials. The duration for the envelope function approaching to zero is 2.1 s and the initial magnetic

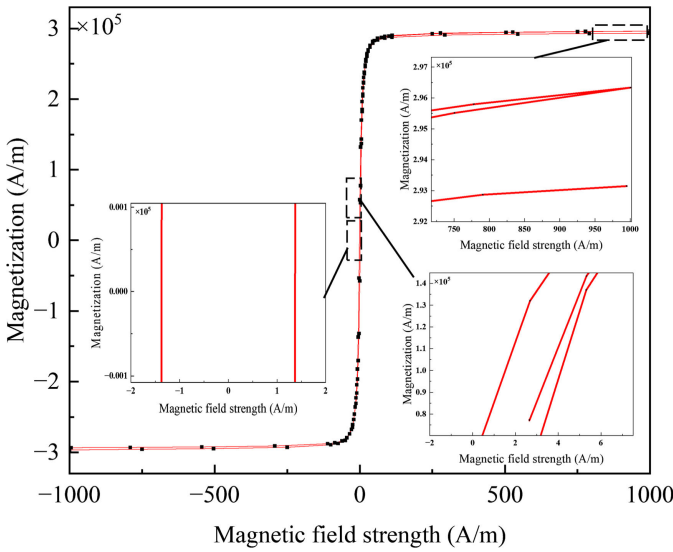


Fig. 3. Measured hysteresis loop of the ferrites used to make magnetic shield.

TABLE II
PARAMETERS OF JA MODEL

Parameter	M_s	α	a	c	k
Value	3.6×10^5 A/m	1.0×10^{-4}	14.1 A/m	0.1	1.16

field is 30 A/m, as shown in Fig. 4(a). The parameters in the JA model are determined when first cycle of demagnetization curve is consistent with the measured initial hysteresis loop, as shown in Table II. By using these parameters, the residual magnetization at the endpoint is calculated. In Fig. 4(b), the initial magnetic field of 30 A/m yields a residual magnetization of 4.56 nT. To analyze the effect of different f and H_0 values on the residual demagnetization, the calculations are repeated, as listed in Table III. It is found that the residual magnetization of endpoint increases with the initial magnetic field H_0 . When the initial magnetic field H_0 is the same, increasing frequency can reduce the residual magnetization. A theoretical magnetic remanence of 0.15 nT is observed in the ferrite material when $f = 50$ Hz and $H_0 = 40$ A/m. These degaussing parameters will be used in the experiments to obtain the best demagnetization.

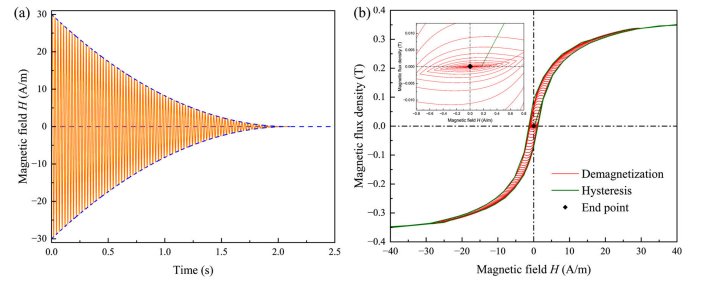
III. SYSTEM DESCRIPTION AND PRELIMINARIES

A. Simulation of Magnetic Field Distribution

To analyze the shielding factor of a cubic magnetic shield, the magnetic field distribution inside the ferrite magnetic shield is simulated. The shielding factor S is defined as a ratio between the external magnetic field B_e and the magnetic field of any point B_o inside the shielding region

$$S = B_e/B_o. \quad (14)$$

In the ANSYS Maxwell software, the simulation model is set to have the same dimension as the actual magnetic shield in Fig. 2(a) and (b). The relative magnetic permeability of

Fig. 4. JA models when $H_0 = 30$ A/m, $f = 40$ Hz, $n = 2$, and $T = 2.1$ s. (a) Demagnetization field with the second-order envelope function. (b) 40 cycles of demagnetization.TABLE III
RESIDUAL MAGNETIZATION UNDER DIFFERENT FREQUENCIES
AND INITIAL MAGNETIC FIELD

Initial magnetic field (A/m)	30 Hz	40 Hz	50 Hz	60 Hz
30	230 nT	4.56 nT	1.15 nT	0.41 nT
40	1270 nT	45.6 nT	0.15 nT	1.98 nT
50	2290 nT	170 nT	3.87 nT	5.57 nT
60	8560 nT	420 nT	2.16 nT	9.17 nT

material used is set to 9000. The ferrite cubic magnetic shield is placed in an $800 (X) \times 800 (Y) \times 800 (Z)$ mm vacuum. By defining the tangential field on the vacuum surfaces, a uniform external magnetic field is applied. Because the cubic magnetic shield is centrally symmetric, the simulation results are the same when the external field comes from the x -, y -, and z -directions. In Fig. 5, an external magnetic field of $1 \mu\text{T}$ is applied in the y -direction, and the residual magnetic field at the center is 0.809 nT, indicating that the ferrite cubic magnetic shield can provide a shielding factor of 1236. In a region of 20 mm around the center, the magnetic field gradient is 0.018 nT/mm in the x -direction, 0.035 nT/mm in the y -direction, and 0.021 nT/mm in the z -direction. In the simulation, the magnetic field is applied in the y -direction, and most of the magnetic flux is contained in the XZ planes, as shown in Fig. 6. This causes the different field distributions in Fig. 5. The results indicate that cubic magnetic shield can provide a low field environment with a small drift.

B. Demagnetization Coils

To realize the minimum residual magnetization in the ferromagnetic materials, several commonly used degaussing coils structures are analyzed using the COMSOL Multiphysics software. A promised degaussing creates an environment with the smallest magnetic field and gradient in the central area. When a dc current follows through the degaussing coils, the magnetic flux density B is introduced to the surfaces of cubic magnetic shield. The magnitude of the magnetic flux density B depends on the structures of degaussing coils. To realize the best residual field, the demagnetization coils need to enclose all edges and holes of the cubic magnetic shield. Sun et al. [33] adopted the L-coils along the edges to construct a magnetic shield with beveled edges and open

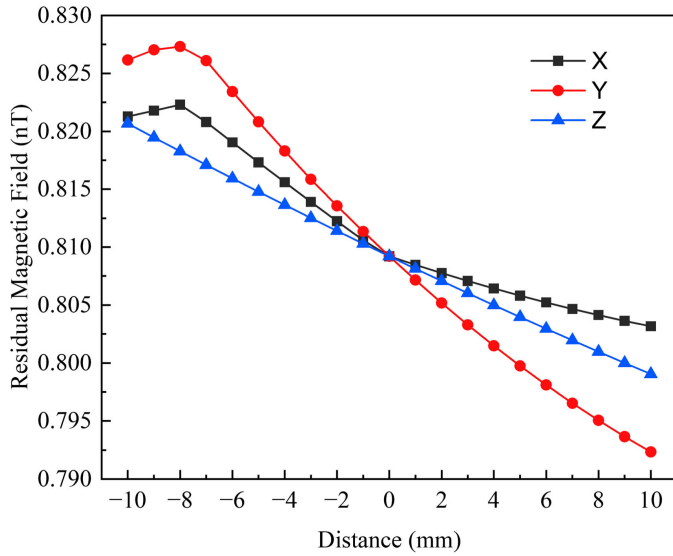


Fig. 5. Simulated results of magnetic field distribution in the x -, y -, and z -directions.

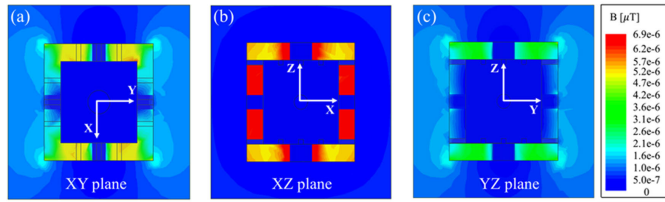


Fig. 6. Magnetic field distribution inside the cubic ferrite magnetic shield. (a) XY plane. (b) XZ plane. (c) YZ plane.

corners, which substantially improve the magnetic shielding factor. In Fig. 7(a) and (c), the magnetic flux density B mainly locates at the corners, where the ferromagnetic materials can easily reach to saturation. However, the less flux density B on the walls of the cubic ferrite magnetic shield will cause the saturated corners come out of saturation later than elsewhere. The insufficient demagnetization cycles on the walls of the cubic magnetic shield can lead to an uneven distribution of the residual magnetization on shield's surface. To solve this problem, more coil windings can be evenly spelled along the walls, as shown in Fig. 7(b) and (d). Under these configurations, the flux density B is evenly distributed on the side walls of the magnetic shield. The configuration of the allocated I-coils yields a much more homogenous magnetic field on the shield's side wall under the same number of degaussing cycles.

After the magnetic flux B on the surface of the magnetic shield is simulated, the central magnetic flux densities under different coil winding structures are analyzed via COMSOL Multiphysics software, as shown in Fig. 8(a)–(d). In the analysis, the magnetomotive force, the degaussing duration time, and the materials parameters, such as M_s , α , a , c , g , and k , are kept the same. The simulation result shows that the L-coils and X-coils have the obvious field gradient in the diagonal directions, which is due to the uneven flux density distribution on their walls and corners, as shown in Fig. 8(a) and (c). Although the allocated L-coils can reduce the

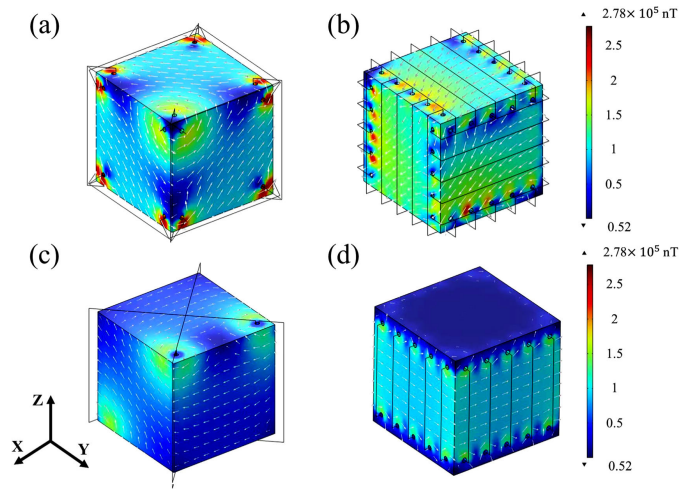


Fig. 7. Simulated magnetic flux density B . (a) L-coils. (b) Allocated L-coils. (c) X-coils. (d) Allocated I-coils.

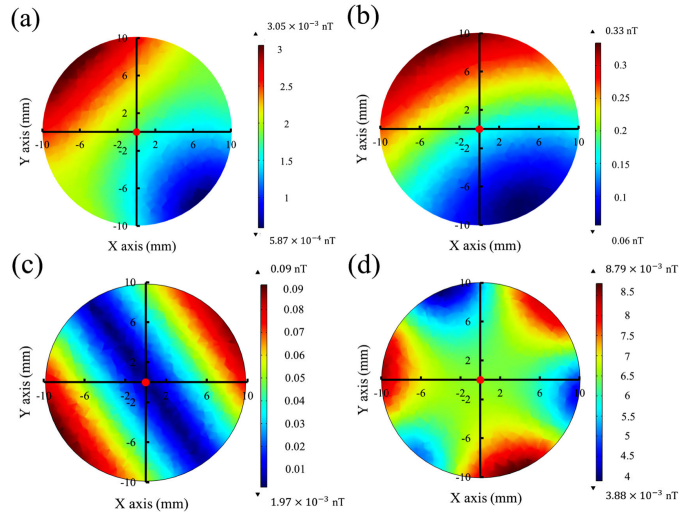


Fig. 8. Flux density B in the central area. (a) L-coils along the edges. (b) Allocated L-coils. (c) X-coils. (d) Allocated I-coils.

magnetic field gradient, the allocated I-coils obviously produce a more homogenous residual magnetic field in the central area, as shown in Fig. 8(b) and (d). In the later experiment, the allocated I-coils are selected as the degaussing coil structure to realize the minimum residual magnetization in the cubic ferrite magnetic shield.

C. Magnetic Noise Measurements

The magnetic noise of the cubic ferrite shield is measured using an atomic SERF magnetometer in a normal inertial measurement state [see Fig. 9(a)]. A cubic Pyrex cell ($8 \times 8 \times 8$ mm), containing a drop of the potassium metal, is injected with 50 torr of N_2 gas and 650 torr of 4He gas. To realize the SERF regime, the vapor cell is placed inside a six-layered μ -metal shielding and our cubic magnetic shield. The remaining environmental magnetic field passing through the magnetic shield is compensated by the field coils around the vapor cell. During the experiment, the potassium

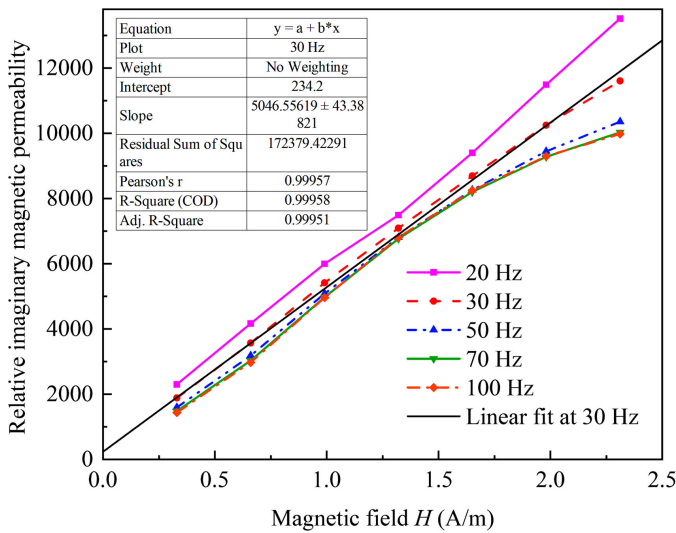


Fig. 10. Measurement results of the relative imaginary parts of the complex magnetic permeability.

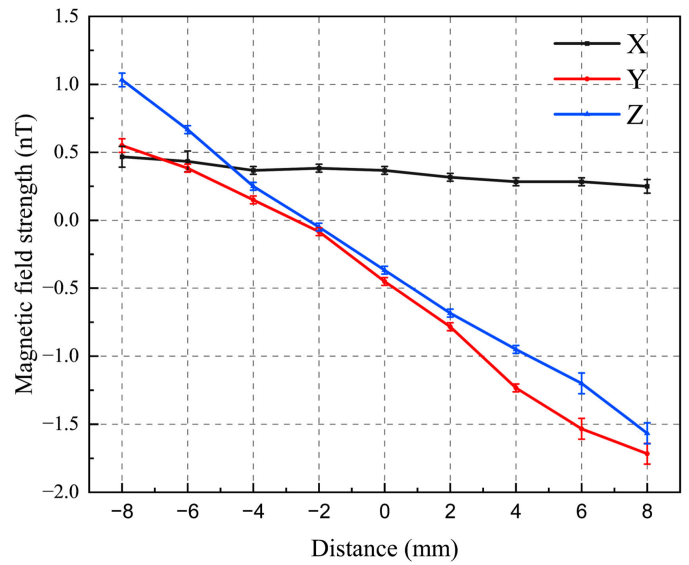


Fig. 12. Measured magnetic field in the x -, y -, and z -directions.

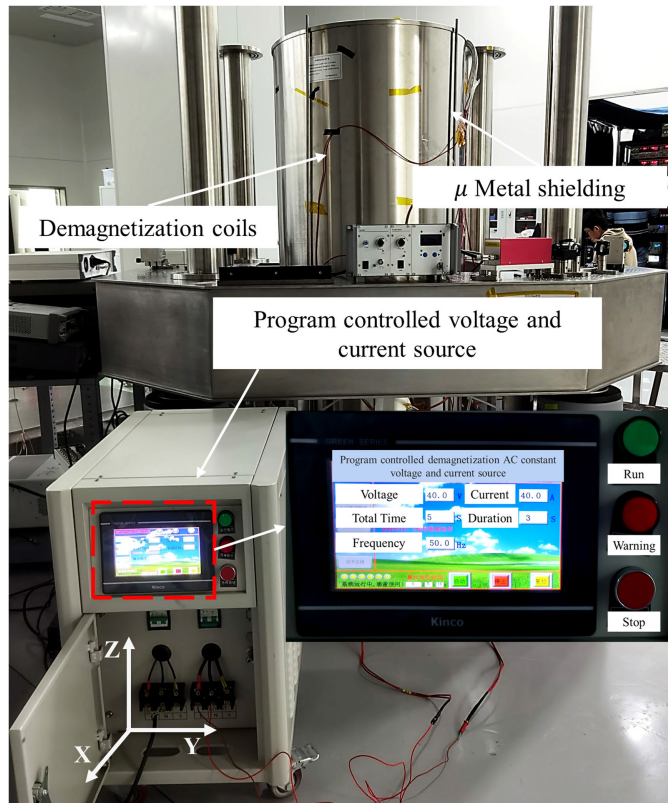


Fig. 11. Experimental setup for magnetic field measurement.

element method (Section II). Without field compensation coils, such cubic ferrite magnetic shield can provide a low magnetic field environment with a small drift for the ultrasensitive atomic magnetometers.

Using the finite element method, the square of the integral magnetic field intensity H^2 over the volume at different points along the central z -axis is analyzed. By substituting the imaginary part of the complex magnetic permeability $\mu'' = 234.2$ and a temperature of 25°C into (7), the magnetic noises of cubic ferrite magnetic shield at different points can

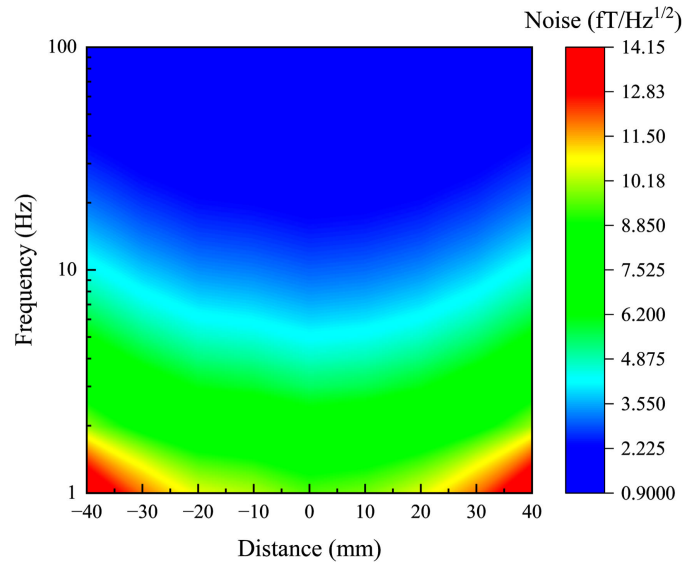


Fig. 13. Contour plot of the calculated magnetic noise of the cubic ferrite magnetic shielding at different positions and frequencies.

be calculated (see Fig. 13). The calculated magnetic noise is nearly symmetric around the center. Due to the central symmetric of the cubic geometry, the magnetic noise distributions should also be the same along the central y - and z -axes. The usefulness of this distribution is that it provides a simple design condition for the compensation and alignment for the traditional field coils [34], [35]. Meanwhile, the magnetic noise increases rapidly when the frequency is below 10 Hz, and the largest magnetic noise occurs at the points of ± 40 mm when the frequency is 1 Hz. Based on the results of finite element method, the averaged magnetic noise is $1.3 \text{ fT/Hz}^{-1/2}$ from 2 to 100 Hz.

Fig. 14 shows the measurement results at points 1–3. Excluding large environmental disturbance at a low frequency, such as microvibration and sensors' $1/f$ noise [36], [37], the magnetic noise spectrum of measurement is analyzed

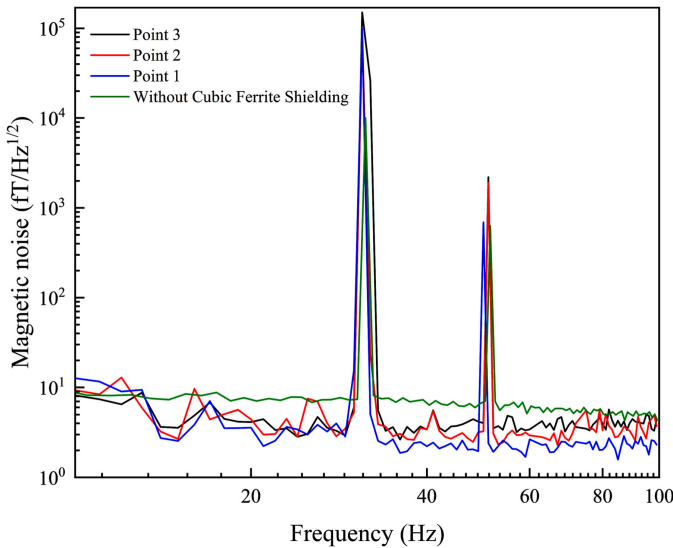


Fig. 14. Noise spectrum of the magnetic field measurement. The green line is the magnetic noise with the cubic ferrite shield. The black line is central point, the blue line is point 1, the red line is point 2, and the black line is point 3.

from 15 to 100 Hz. In Fig. 14, the rapid increment of the magnetic noise at 30 Hz is due to the given calibration signals in the experiment. Also, a large noise occurs at 50 Hz because of the frequency of the main electricity at 50 Hz. The result shows that the magnetic noise at point 1 (center) is smaller than the other two points and the magnetic noise increases with the distance from the central point, which is consistent with the calculated results in Fig. 13. After excluding the magnetic noise of calibration at 30 Hz and of the main electricity at 50 Hz, the averaged magnetic noise at the ferrite magnetic shield's center is $2.62 \text{ fT/Hz}^{-1/2}$ from 20 to 100 Hz. When there is no cubic ferrite magnetic shield, averaged magnetic noise is $6.12 \text{ fT/Hz}^{-1/2}$ from 20 to 100 Hz. Compared with the noise spectrum of permalloy shield, the noise level inside cubic ferrite magnetic shield is reduced more than 133%. It is found that the magnetic noise measurements are limited by the sensitivity of the SERF magnetometer used. Noise measurements less than $1 \text{ fT/Hz}^{1/2}$ could be realized by using a more sensitivity magnetometer. These experimental results all indicate that the cubic ferrite magnetic shield can significantly reduce the noise level.

The noise spectrum of the magnetic field measurement is shown in Fig. 14. The green line is the magnetic noise with the cubic ferrite shield. The black line is central point, the blue line is point 1, the red line is point 2, and the black line is point 3.

V. CONCLUSION

In sight to suppress the eddy current magnetic noise of MSR made of permalloy, a cubic Mn–Zn ferrite magnetic shield is proposed. The JA demagnetization model is used to analyze the theoretical minimum residual magnetization in the ferrite materials. The degaussing effects of four different degaussing coil structures are analyzed, and the allocated I-coils produce a homogenous residual magnetic field in the central area. After

applying the optimized demagnetization frequency of 50 Hz and an initial applied field of 40 A/m, the residual field of the cubic magnetic shield was measured. Within a distance of $\pm 10 \text{ mm}$ near the center, the residual magnetic field is below 0.5 nT at the x -direction and is below 1.72 and 1.57 nT at the y - and z -directions, respectively. Without any magnetic compensation coil, the ferrite magnetic shield can provide a low magnetic field environment. Meanwhile, the magnetic noise of the cubic magnetic shield was analyzed using the FEM. The calculation results show the magnetic noise distribution of cubic ferrite magnetic shield is symmetric, which is convenient for the field compensation and alignment in the coil design. The actual magnetic noise of cubic ferrite magnetic shield was measured by using an SERF magnetometer. After excluding the magnetic noise of calibration at 30 Hz and of main electricity at 50 Hz, the measured magnetic noise at the center is $2.62 \text{ fT/Hz}^{-1/2}$ from 20 to 100 Hz. Compared with the noise spectrum of permalloy shield, the noise level inside cubic ferrite magnetic shield is reduced more than 133%.

This study proposed an insight to reduce the magnetic noise for traditional MSRs. With the help of cubic ferrite shield, the magnetic noise of MSRs can be significantly reduced. By assembling multiple ferrite panels together, it is possible to create a larger shielding space. The cost of scaling up depends on the number of ferrite panels used. It is practical to create a larger shielding room by using the ethyl adhesive to stick more ferrite panels together. The durability of the cubic ferrite magnetic shield can last for several years, depending on the degrading period of ferrite materials. Due to the fragility of ferrites materials, large loads on the shields should be avoided in order to prevent any damages. To further investigate the noise characteristics of the ferrite magnetic shield, magnetometers with the higher sensitivity are required.

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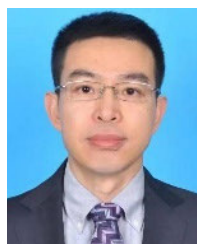
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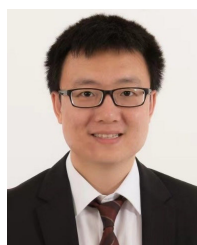
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